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Document Owner: VP, Core Networks

Service Coverage and Infrastructure Footprint Document: VI Business (India)

1) Executive Purpose and Applicability

The Service Coverage and Infrastructure Footprint Document is the definitive internal reference for VI Business (India)'s physical and logical network reach. It establishes the exact scale of terrestrial fiber, wireless boundaries, and data center locations to ensure Bid Teams do not overcommit resources during RFP negotiations.

Intended Audience: Network Planners, Bid Authors, Pre-Sales Architects. **Applicability:** For addressing RFP sections regarding "Geographic Coverage", "Last-Mile Reach", "Data Residency", and "High-Availability Edge".

2) Definitions and Assumptions

Definitions:

- **DIA:** Dedicated Internet Access (Enterprise Broadband).
- **NLD:** National Long Distance (Fiber Routes).
- **RoW:** Right of Way.

Assumptions:

- **Internal Control:** Terrestrial coverage maps frequently shift. Blanket RFP commitments citing active fiber at specific local ZIP codes require physical feasibility verification by the Network Operations team before executing the MSA.

3) Detailed Core Sections

3.1 Core Terrestrial Network Reach VI Business operates a sovereign national backbone consisting of 250,000+ km of domestic optical fiber. This infrastructure interconnects our 12 major Tier-III data centers and extends to urban business districts and industrial corridors across all 28 states and 8 Union Territories.

3.2 Wireless Network Coverage (LTE/5G and IoT)

- **Mobile RAN:** Providing 98%+ population coverage natively. This macro network acts as the primary failover path (via 4G/5G Business LTE), ensuring our 99.99% Hub SLA remains robust even if terrestrial fiber routes experience physical severing.
- **IoT Coverage:** The footprint seamlessly supports 2.5 million+ remote edge sensors and logistics trackers. Endpoint traffic is isolated via our dedicated Custom APNs (cAPN) to prevent public internet exposure.

3.3 Data Center and Sovereign Nodes

- **Data Center Fabric:** 12 major Tier-III facilities located nationwide.
- **SOC Command Centers:** Primary ITIL v4 certified Security Operations Center (SOC) operates from Pune, Maharashtra, providing geographically redundant fallback to the Mumbai facility.
- **Sovereign Compliance (DPDP Act 2023):** 100% telemetry data and SIEM log data localization inside India. No routing tables, payload inspections, or syslog traffic traverse international egress gateways without explicit client (Data Fiduciary) and DoT authorization.

3.4 Cloud On-Ramp Express Interconnects To bypass standard ISP latency variables, VI Business operates dedicated hyperscaler Meet-Me-Rooms (MMR) anchored primarily within our NESCO IT Park, Mumbai DC and Chennai DC:

- **Microsoft Azure ExpressRoute:** Dedicated 1 Gbps to 10 Gbps Active-Active private peering circuits terminating directly into Azure Central India.
- **AWS Direct Connect:** Specialized Layer 2/3 interconnects supporting AWS ap-south-1.
- *Performance Baseline:* These native integrations secure business-critical routing (e.g., SAP S/4HANA, core banking transactions) consistently beneath `< 8ms` round-trip latency to the hyperscaler edge.

4) Evidence and Proof-Artifact Map

Infrastructure Scope	Validation Material	Repository Location
Nationwide RAN Map	`VI_LTE_NLD_Footprint.kml`	`SharePoint/VIB_Network_Topology`
Data Residency Compliance	`DataLocality_Compliance_Audit.pdf`	`SharePoint/VIB_Compliance_Vault`

5) Governance and Ownership

- **Owner:** VP, Core Networks
- **Reviewers:** VP Engineering, Enterprise CISO.
- **Review Frequency:** Bi-Annually.

6) Constraints, Exclusions, and Dependencies

- **Constraints:** Hardware replacement slates (e.g., 4-hour NBD dispatch) at remote Tier-3 locations heavily depend on proximity to designated regional engineering depots mapped during the site survey phase.
- **Exclusions:** Global cross-border international links (e.g., Mumbai to London IPLC) operate under distinct contractual rules and international carrier SLAs outside this standard domestic policy.